

✓ Heap Sort Implementation

```
import time

def heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2

    if left < n and arr[left] > arr[largest]:
        largest = left

    if right < n and arr[right] > arr[largest]:
        largest = right

    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]

        heapify(arr, n, largest)

def heap_sort(arr):
    n = len(arr)

    for i in range(n // 2 - 1, -1, -1):
        heapify(arr, n, i)

    for i in range(n - 1, 0, -1):
        arr[0], arr[i] = arr[i], arr[0]

        heapify(arr, i, 0)

print("HEAP SORT")

n = int(input("Enter the number of elements: "))

arr = []

print("Enter", n, "elements:")
for i in range(n):
    element = int(input())
    arr.append(element)
start_time = time.perf_counter()

heap_sort(arr)

end_time = time.perf_counter()

execution_time = end_time - start_time

print("\nSorted Array:", arr)

print("\nExecution Time: {:.10f} seconds".format(execution_time))

print("\nTime Complexity:")
print("Best Case : O(n log n)")
print("Average Case : O(n log n)")
print("Worst Case : O(n log n)")

print("\nSpace Complexity: O(1)")
```

```
HEAP SORT
Enter the number of elements: 7
Enter 7 elements:
6
2
4
1
9
0
4

Sorted Array: [0, 1, 2, 4, 4, 6, 9]
```

Execution Time: 0.0001032170 seconds

Time Complexity:

Best Case : $O(n \log n)$

Average Case : $O(n \log n)$

Worst Case : $O(n \log n)$

Space Complexity: $O(1)$

max heap sorting

```
import time
def max_heapify(arr, n, i):
    largest = i
    left = 2 * i + 1
    right = 2 * i + 2

    if left < n and arr[left] > arr[largest]:
        largest = left

    if right < n and arr[right] > arr[largest]:
        largest = right

    if largest != i:
        arr[i], arr[largest] = arr[largest], arr[i]

        max_heapify(arr, n, largest)

def heap_sort(arr):
    n = len(arr)

    for i in range(n // 2 - 1, -1, -1):
        max_heapify(arr, n, i)

    for i in range(n - 1, 0, -1):
        arr[0], arr[i] = arr[i], arr[0]

        max_heapify(arr, i, 0)

print("MAX HEAP SORT")

n = int(input("Enter the number of elements: "))

arr = []

print("Enter", n, "elements:")
for i in range(n):
    arr.append(int(input()))

print("\nOriginal Array:", arr)

start_time = time.perf_counter()

heap_sort(arr)

end_time = time.perf_counter()

execution_time = end_time - start_time
print("Sorted Array:", arr)

print("\nExecution Time: {:.10f} seconds".format(execution_time))

print("\nTime Complexity:")
print("Best Case :  $O(n \log n)$ ")
print("Average Case :  $O(n \log n)$ ")
print("Worst Case :  $O(n \log n)$ ")

print("\nSpace Complexity:  $O(1)$ ")
```

MAX HEAP SORT

Enter the number of elements: 8

Enter 8 elements:

6

8

0

1

4
3
2
7

Original Array: [6, 8, 0, 1, 4, 3, 2, 7]
Sorted Array: [0, 1, 2, 3, 4, 6, 7, 8]

Execution Time: 0.0001335970 seconds

Time Complexity:

Best Case : $O(n \log n)$

Average Case : $O(n \log n)$

Worst Case : $O(n \log n)$

Space Complexity: $O(1)$